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ON-DELAY AND OFF-DELAY TIMERS
Using CD4050 And CD4049

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Many devices, such as water pumps, exhaust fans, and geysers, need to run for only a limited period. Forgetting to switch them off can result in unnecessary power consumption and wear. This timer system can automatically switch off a garage light after a car is parked or turn off an exhaust fan, water pump, or outdoor light after a preset time, reducing power consumption and wear.

Some applications also require a device to remain off for a few minutes after the power supply is restored. For example, a refrigerator or air-conditioner compressor should preferably not restart immediately. This allows the refrigerant pressure to stabilise and helps protect the compressor from excessive stress caused by rapid switching.

Two simple timer circuits are described here. The first is an on-delay timer that switches on the connected appliance after a preset delay. Fig. 1 shows the author's on-delay timer prototype assembled on a breadboard.

The second is an off-delay timer, which keeps the appliance on for a preset period and then switches it off. Fig. 2 shows the author's prototype of the off-delay timer assembled on a breadboard.

Both circuits use a minimum number of components and are easy to assemble. The delay can be varied continuously from approximately three to fifteen minutes by adjusting potmeter VR1.

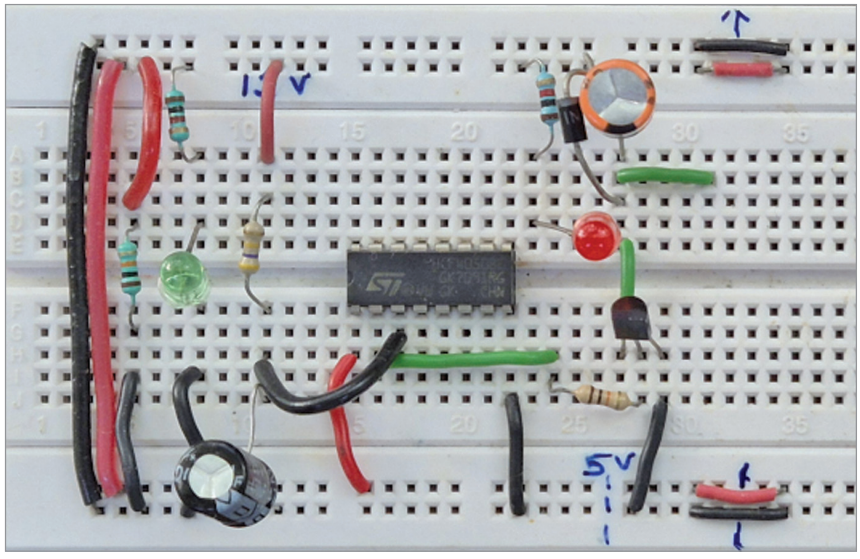


Fig. 1: Author's prototype of the on-delay timer

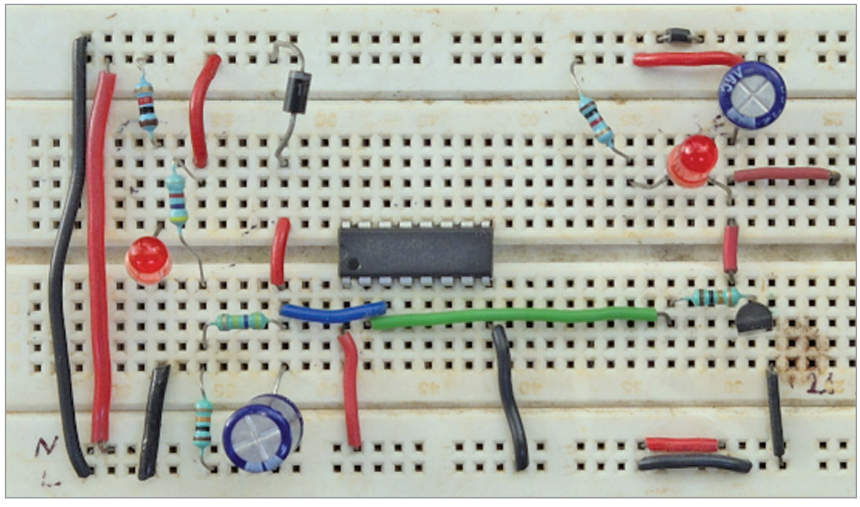


Fig. 2: Author's prototype of the off-delay timer

Circuit and working

Two types of timers are described here: an on-delay timer and an off-delay timer.

The on-delay timer is built around CD4050 (IC1), a non-inverting hex buffer. The IC contains six buffers, but only IC1A is used in this system.

When the system is powered on, capacitor C1 is initially discharged. C1 starts charging through resistor R2 and potentiometer VR1. The voltage across C1 gradually increases. When it reaches the input threshold of the CD4050, its output goes high. This drives transistor T1, which energises relay RL1. The appliance connected through CON2 is then switched on.